

Historical Data Preservation Policy

Document purpose: Explains how old (pre-migration) information is handled throughout the repository — what was removed, what was preserved, and how preserved content is clearly marked so it cannot be accidentally cited as current.

1. Background

During Phases 8–10 the project underwent a major **nomenclature migration** (see [nomenclature_migration.md](#) (nomenclature_migration.md), DEC-011/DEC-012):

Item	Old (v1)	Current (v2)
P0 baseline	Index-based round-robin (uniform_allocation)	Spatially-stratified latitude selection (spa-tially_stratified_allocation)
P0 mean RT (K = 20)	~18.5 min (optimisation), ~8.08 min (simulation)	~4.75 min (optimisation), ~3.17 min (simulation)
Firehouse capacity default	5 units	2 units
P0 8-min coverage (K = 20)	~64 %	~99.7 %

The migration touched **38 files** across configs, source code, results, notebooks, and documentation (see [docs/project_wide_audit_report.md](#)).

2. Strategy: Remove vs. Preserve

2.1 Content that was REMOVED (stale / incorrect)

These artefacts were deleted or overwritten because they would mislead if encountered without context:

Category	Action	Example
Stale notebook outputs	Cleared cell outputs	07_production_results.ipynb — 16 cells showed old P0 = 8.08 min
Incorrect metric labels	Replaced in-place	policy_comparison.csv — P0 label changed from “Uniform” → “Spatially-Stratified Uniform”
Wrong config defaults	Updated	configs/optimization.yaml — firehouse_capacity: 5 → 2
Old P0 RT values in authoritative docs	Replaced	technical_report.md , executive_summary.md , optimization_results.md
Dead code paths	Not removed, but deprecated with runtime warning	uniform_allocation() now emits DeprecationWarning

2.2 Content that was PRESERVED (with clear markers)

Some old information has legitimate historical value — it documents why decisions were made and provides an audit trail:

Document	Old content retained	How it is marked
docs/ policy_tradeoff_analysis.md	Full analysis using original index-based P0	 Historical Context Note banner at top of file
docs/nomenclature_migration.md	Complete before/after comparison of old vs new P0	Entire document is the migration record
docs/decisions_log.md (DEC-011, DEC-012)	Rationale for replacing old P0	Entries clearly dated and labelled “NEW”
docs/assumptions_log.md (A14, A15)	Reference to deprecated original P0	Labelled “(Internal Reference)” and links to migration docs
docs/conceptual_model_selection.md	Mentions “original P0” RT of 8.08 min	In narrative context explaining the migration rationale
docs/final_summary.md	“8.08 → 3.17 min” improvement narrative	Presented as a before→after comparison
docs/notebook_audit_report.md	Audit log documenting what was changed	Meta-document describing the audit itself
docs/project_workflow_wbs.md	Task entry for P0 spatial analysis	Historical task log
src/ems_readiness/optimization/policies.py	<code>uniform_allocation()</code> function body	<code>DeprecationWarning</code> emitted at runtime; docstring says DEPRECATED
src/ems_readiness/optimization/allocator.py	<code>baseline_uniform()</code> method	Status = “Baseline (deprecated)”; docstring recommends <code>baseline_p0()</code>

2.3 Result files with mixed historical data

File	Contains old data?	Status
results/production_v2/simulation/all_results_raw.csv	Yes — contains K = 10 rows where P0 RT > 4 min	Expected — small fleet sizes legitimately have higher RT
results/tables/posthoc_comparisons.csv	References to old statistical tests	Legacy from v1 production run; superseded by <code>results/production_v2/tables/</code>
results/optimization/policy_comparison.csv	No old data	✓ Clean — all P0 labels are “Spatially-Stratified Uniform”

3. Marking Conventions

3.1 Documentation banners

Historical documents that retain old data carry a blockquote banner:

```
> ** ⚠ Historical Context Note:** This analysis uses the **original  
> index-based uniform P0** (deprecated), not the current spatially-stratified  
> P0 baseline. See [`docs/nomenclature_migration.md`](nomenclature_migration.md)  
> for the full nomenclature history.
```

3.2 Code deprecation

```
warnings.warn(  
    "uniform_allocation() is deprecated as the P0 baseline. "  
    "Use spatially_stratified_allocation() instead.",  
    DeprecationWarning, stacklevel=2,  
)
```

3.3 Config comments

```
# Capacity sensitivity analysis (docs/capacity_sensitivity_analysis.md)  
# shows cap=2 matches or improves upon cap=5 at K≤40.  
firehouse_capacity: 2
```

4. Safeguards Against Accidental Use

Safeguard	Location	Mechanism
Runtime deprecation warning	<code>policies.py</code>	<code>DeprecationWarning</code> on <code>uniform_allocation()</code>
Docstring flags	<code>policies.py</code> , <code>allocator.py</code>	“DEPRECATED” in first line of docstring
Verification script	<code>scripts/verify_project_consistency.py</code>	Automated 41-check scan for stale metrics, labels, and config values
Deprecated function registry	Verification script §5	Lists all deprecated functions and their replacements
JSON verification report	<code>results/consistency_verification_report.json</code>	Machine-readable audit trail
Historical banner requirement	This document §3.1	All docs with old data must carry  banner
Nomenclature migration guide	<code>docs/nomenclature_migration.md</code>	Canonical mapping of old → new names

5. Deprecated Functions — DO NOT USE

Function	File	Replacement
<code>uniform_allocation()</code>	<code>src/ems_readiness/optimization/policies.py</code>	<code>spatially_stratified_allocation()</code>
<code>baseline_uniform()</code>	<code>src/ems_readiness/optimization/allocator.py</code>	<code>baseline_p0()</code>

These functions remain in the codebase for backward-compatibility and historical traceability only. They emit `DeprecationWarning` at runtime and are excluded from all production scripts and notebooks.

6. Verification Checklist

Before any submission or packaging, run:

```
python scripts/verify_project_consistency.py
```

The script checks:

- [] All configs have correct parameters (capacity = 2, correct policy names)
- [] All result files have correct metrics (P0 ~3-5 min, not 8+)
- [] All scripts use correct functions (no stale `uniform_allocation` calls)
- [] All authoritative docs have correct nomenclature
- [] No stale data references in production results
- [] All historical docs have warning banners

Exit code 0 = all critical checks pass.

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